

EnviroSense: AI-Driven Microclimate Control for Sustainable Agriculture Using Edge Computing

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Abstract. The application of AI-driven microclimate control is vital to enhancing agricultural output and minimizing resource wastage. This paper aims to present how real-time IoT sensor data, machine learning, and mobile edge computing can be leveraged to improve crop yield prediction and promote sustainable farming practices. The system developed in this paper achieves 98% accuracy in field condition classification, providing valuable insights for farmers while addressing key challenges in the agricultural sector, such as climate variability, resource scarcity, and precision agriculture. It incorporates machine learning algorithms, real-time sensor data, and a user-friendly mobile application that enables farmers to manage their crops efficiently, predict yields, and optimize resource allocation. Some of the major features include role-based access control, predictive analytics, and offline functionality, ensuring usability even in low-connectivity environments. The integration of interactive data visualizations enhances real-time decision-making by presenting trends and predictions in an intuitive format. Using the Random Forest model, this system can label field conditions with an accuracy of 98%, significantly improving crop health monitoring and reducing prediction errors. Additionally, the model leverages precision agriculture techniques to provide tailored recommendations based on environmental conditions. Some challenges include sensor noise and computational complexity in certain models, which are mitigated through data preprocessing techniques. This work directly addresses inefficiencies in agriculture and contributes to reducing environmental degradation through the application of AI-based microclimate management. Future improvements will focus on integrating real-time IoT sensors, expanding crop variety support, and enhancing mobile edge computing capabilities for more robust and scalable precision agriculture solutions.

Keywords: Microclimate Control, Mobile Edge Computing, AI-driven Climate Control, Sustainable Farming, IoT, Crop Yield

1. Introduction

There is great potential for the agricultural sector to increase its output and become more efficient by addressing issues regarding resources, climate change, and the requirements for sustainable production systems. The provision of technology to

small and medium-scale farms is crucial in enhancing the efficiency and sustainability of farming. This challenge can be solved with microclimate monitoring, with the use of machine learning models, and with mobile edge computing.

By integrating real-time sensor data and artificial intelligence insights into a single intuitive mobile application the farmers are provided with recommendations on how to enhance crop health and yield with the use of resources that would otherwise be inefficient.

With the help of edge computing affordable sensors, and predictive analytics, farmers are able to make proper decisions. They offer important insights into how to decrease inefficiency and increase sustainability through making interventions using artificial intelligence and the Internet of Things in agriculture. This integration demonstrates the possibilities for the shift-type advancements which present the list of important insights into the areas of increased productivity and sustainable performance.

In this study, we propose the EnviroSense system that integrates real-time sensor data machine learning models, and mobile edge computing to address the inefficiencies in modern agriculture. It deploys affordable sensors to gather information on temperature, humidity, and soil pH among other environmental conditions. It then processes the data by subjecting it to machine learning models (in this case Random Forest) for field condition categorization and yield estimation. This framework provides intelligent graphics and valuable information even in the absence of well-developed infrastructure for connection. By focusing on the cost and the ability of the system to scale down to suit small and medium-scale farmers, this study presents a new way of addressing the issue of equipping farmers with the right technologies for sustainable and efficient farming.

In this paper, the authors describe the EnviroSense system, the details of its implementation, technical problems, the system architecture, the integration of AI, and experimental evaluation. The authors also describe the possibilities for the expansion of the study and its future directions. This study adds to the body of knowledge on precision agriculture by offering a feasible and scalable solution that bridges the gap between the availability of modern agricultural technologies and the ability of small-scale farmers to access them. With the help of edge computing, EnviroSense is able to support real-time, offline operations thus reducing latency and the need for reliance on the cloud. Also, it shows how AI and IoT can be harnessed in managing resources and improving production and provides a roadmap for the implementation of sustainable agricultural technologies in the areas of low income.

This paper is organized as follows: Section 2 provides background information on AI, IoT, and Edge Computing in agriculture, discussing their role in microclimate control and sustainable farming while highlighting challenges in integrating modern technology with agriculture. Section 3, titled Problem Definition and Design Considerations, identifies key agricultural challenges and explains the design principles behind the proposed solution, including scalability, affordability, and offline functionality. Section 4, Engineering Solutions, details the system architecture, covering development phases, hardware and software design, and system implementation, along with data flow and processing methods. Section 5, AI Integration, discusses the machine learning models used, their selection criteria, preprocessing techniques, and implementation details for field condition classification

and yield prediction. Section 6, Model Evaluation and Performance Analysis, presents the experimental results, including model accuracy, feature importance analysis, and comparative evaluations of different models. Section 7, Key Findings and Conclusion, summarizes the primary outcomes of the research, emphasizing the effectiveness of the EnviroSense system in improving crop monitoring and predictive analytics. Finally, Section 8, Future Work, outlines areas for further research, including expanding crop varieties, integrating additional AI-driven analytics, and improving system scalability for broader adoption.

1.1 Background

Climate changes, wastage of resources, and the demand for sustainable livelihoods put an added burden on agriculture. Recent advancements in AI and IoT give promising results in solving these problems by implementing precise monitoring and control over environmental conditions using affordable technologies. Studies show that AI models integrated with IoT systems improve yield prediction accuracy and optimize irrigation scheduling, contributing to sustainable agricultural practices [1]. Research shows that the integration of AI and IoT in agriculture significantly improves resource efficiency, particularly in the management of water and fertilizers, enhances productivity, and reduces waste. Studies in [1] and [2] highlight that AI-driven irrigation systems can optimize water usage by 30%, while IoT-based monitoring has improved fertilizer efficiency by 25%. Recent advancements in AI and IoT have significantly contributed to the development of smart agriculture systems, enhancing precision farming and resource management[3].

Microclimate control is one of the most important features in modern agriculture, allowing the observation of essential environmental parameters such as temperature, humidity, and soil quality [4]. However, small and medium-scale farms are often excluded from accessing such advanced technologies due to their high cost and complexity. This prevents them from optimizing crop health and adopting sustainable farming practices.

Small-scale farmers are crucial for global food security, contributing significantly to agricultural production and rural economies. Beyond food provision, they offer environmental, social, cultural, and economic benefits, such as enhancing crop diversity, providing employment, and promoting self-sufficiency. However, these farmers face numerous challenges, including limited access to technology, information, markets, and labor shortages [5][6].

Technological advancements in AI and the IoT have revolutionized various sectors, including agriculture. AI enables data-driven decision-making, while IoT facilitates real-time monitoring through interconnected devices. The integration of AI and IoT in agriculture, known as “smart farming”, enhances productivity, optimizes resource use, and promotes sustainable practices [1]. Recent advancements in AI and IoT have demonstrated improvements in crop yield prediction and irrigation optimization, fostering more sustainable agricultural practices. [7]

Despite these advancements, small-scale farmers often struggle to adopt such technologies due to high costs, lack of technical expertise, and infrastructure limitations. Additionally, managing microclimates—localized atmospheric zones with distinct climatic conditions—poses significant challenges. Microclimates can affect crop yields, pest populations, and water availability, necessitating tailored farming

practices to optimize outputs. Traditional methods of microclimate management are labor-intensive and may not provide the precision required for optimal results [8].

Addressing these challenges requires innovative solutions that are accessible, affordable, and user-friendly for small-scale farmers. By leveraging AI and IoT, it is possible to develop systems that provide real-time data and predictive analytics, enabling farmers to make informed decisions about irrigation, fertilization, and pest control. Such technologies can enhance resource efficiency, increase yields, and reduce environmental impacts, thereby contributing to the sustainability and resilience of small-scale farming operations.

The EnviroSense project provides a low-cost sensor-driven solution for microclimates in real time and then integrates AI and mobile edge computing, enabling farmers to embark on informed decision-making that addresses productivity as well as environmental concerns. This study provides the implementation challenges and details of the system to make technology more accessible to small-scale farmers. The system uses random forest, logistic regression, and linear regression models for yield prediction, field condition monitoring, and recommendations.

2. Related Work

The integration of AI and the IoT has significantly transformed precision agriculture. IoT sensors are employed to monitor environmental factors such as temperature, humidity, and soil moisture, while AI algorithms analyze this data to optimize farming practices [9]. IoT sensor networks are essential for enabling real-time data collection, contributing to precision agriculture by enhancing efficiency and reducing resource wastage[10]. In [9], the authors demonstrated that AI-driven climate control systems in polyhouse agriculture enhanced crop quality and improved resource efficiency. These advancements contribute to sustainable farming practices by reducing waste and optimizing resource use. The study also highlights barriers for small-scale farmers, including the high cost of such technologies and the need for specialized technical knowledge to implement them effectively [6].

Edge computing has gained importance in smart farming as it processes data closer to its source, reducing latency and improving the responsiveness of decision-making processes [11]. This is particularly critical for time-sensitive operations, such as irrigation scheduling and pest control. In [12], the authors proposed a framework for integrating edge computing into agricultural systems, demonstrating how it minimizes dependency on cloud infrastructure and reduces operational costs. Their results indicate that edge-based systems are suitable for low-connectivity regions, enabling efficient resource management and real-time decision-making.

The combination of mobile edge computing and IoT, often referred to as the Internet of Agricultural Things, has revolutionized farming operations [9]. By integrating IoT sensors and edge processing capabilities, real-time monitoring and control of critical agricultural processes are achieved. Studies, such as [4], systematically reviewed this integration and highlighted its scalability and adaptability for diverse farming environments. In particular, their findings showed a 20% improvement in resource utilization and a significant increase in crop yields under real-world conditions. Such studies provide a foundation for further exploring the role of mobile edge computing in agricultural innovation.

Microclimate prediction plays a vital role in agricultural planning and management. Advanced AI models, such as DeepMC, are capable of achieving over 90% accuracy in predicting microclimate parameters, leveraging IoT-driven smart farming techniques [9]. The authors used hyper-local environmental data to create predictions that support tailored irrigation scheduling and pest management strategies. Their results emphasize the importance of integrating microclimate predictions with actionable recommendations, thereby improving crop resilience and reducing climate-related risks.

IoT-based systems have also been developed to monitor and manage greenhouse environments, leveraging sensors and actuators to maintain optimal growing conditions. For instance, [13] demonstrated the use of IoT-driven systems to automate environmental control in greenhouses, resulting in a 25% reduction in resource usage and a corresponding increase in productivity. Additionally, their study highlighted the potential of these systems to lay the groundwork for automated, labor-efficient farming practices [12].

Building upon these existing works, the EnviroSense project addresses several challenges in agricultural technology by integrating AI, IoT, and mobile edge computing into a single, cost-effective solution. Unlike previous studies, EnviroSense emphasizes affordability, offline functionality, and intuitive user interfaces to empower small- and medium-scale farmers. This combination of features makes the system particularly impactful for underserved farming communities, addressing both productivity concerns and environmental sustainability [14].

3. Problem Definition and Design Considerations

Modern agriculture faces significant challenges, including resource scarcity, unpredictable weather conditions, and the need for sustainable practices. While advances in artificial intelligence and the Internet of Things have revolutionized precision agriculture, the high cost and complexity of these technologies often limit their adoption by small and medium-sized farms. Effective microclimate management, which is critical for optimizing crop health and resource use, presents a particular challenge.

The EnviroSense system is designed to address these challenges by providing a cost-effective, scalable, and user-friendly solution that integrates AI, IoT, and edge computing. By enabling real-time monitoring and predictive analytics for environmental parameters such as temperature, humidity, and soil pH, the system empowers farmers with actionable insights to optimize resource use and enhance productivity. Offline functionality and role-based access ensure usability across diverse user groups, even in areas with limited connectivity. EnviroSense employs local caching mechanisms and asynchronous data synchronization to handle low-connectivity environments effectively. When internet connectivity is available, sensor data is uploaded to the cloud for analysis. In offline mode, the system continues to process and store data locally using edge computing devices, ensuring uninterrupted operation. Farmers can access the most recent data, with freshness indicators highlighting whether insights are derived from real-time data or cached records. This dual-mode operation enables farmers to maintain operational efficiency and receive actionable insights, regardless of connectivity status. By integrating offline capabilities, EnviroSense empowers farmers in remote areas to make

data-driven decisions without constant internet access. This approach not only reduces downtime but also enhances resilience against connectivity disruptions, ensuring continuous monitoring and intervention. [15] provides a comprehensive overview of technological adaptations through AI, IoT, and robotics in modern farming. Their findings highlight how integrating these technologies can address critical issues faced by small-scale farmers, such as resource scarcity and labor shortages. The EnviroSense system aligns with these insights by leveraging AI and IoT to enhance productivity and efficiency, bridging the technological gap for small-scale farmers.

The development of the EnviroSense system is guided by the following design principles.

Scalability: The architecture supports multiple users, diverse crop types, and large datasets without compromising performance, ensuring adaptability for various farming operations.

Affordability: Low-cost sensors and open-source technologies were selected to lower the barrier to entry for small-scale farmers, making advanced agricultural technology more accessible.

Offline Functionality: Local caching mechanisms were implemented to ensure uninterrupted operation in regions with limited internet connectivity, enabling farmers to access critical data at all times.

User-Centric Design: The system's interface provides role-based access to ensure farmers and administrators receive relevant insights, simplifying data interpretation and decision-making processes.

In [15] the challenges small-scale farmers encounter are reviewed, such as limited market access, climate variability, and resource inefficiencies. Their analysis suggests that AI-driven systems and IoT can mitigate these challenges by offering real-time data analytics and predictive insights. EnviroSense adopts a similar approach, providing affordable sensor-driven solutions that address the unique needs of small-scale farmers. These considerations aim to deliver an inclusive and practical solution tailored to the unique needs of small-scale farmers, contributing to global food security and environmental sustainability.

4. Engineering Solution

4.1 Development Phases and Timeline

To systematically address the problem and implement the EnviroSense system, the development process is structured into four key phases, each with specific milestones and iterative feedback loops to refine features and overcome challenges.

Phase 1: Research and Design

- 1) Conducted a comprehensive review of existing agricultural technologies to identify gaps and limitations in current solutions for small-scale farmers.
- 2) Developed initial architecture and wireframes tailored to meet the unique needs of small-scale farming operations.
- 3) The insights gained during this phase shaped the foundational structure of the system, ensuring scalability and ease of integration with IoT-based sensors.

Phase 2: Model Training and Evaluation

- 1) Machine learning models were trained using synthetic datasets to classify field conditions and predict crop yields.
- 2) Random Forest models were selected and fine-tuned to strike a balance between computational efficiency and predictive accuracy.
- 3) The AI models were iteratively improved based on performance metrics, ensuring robust field condition classification.

Phase 3: Data Integration and Visualization

- 1) IoT-based sensors were deployed to collect environmental data in real time, feeding directly into the system's analytical pipeline.
- 2) The frontend and backend were developed in parallel using React Native for a responsive user interface and cloud-based databases for secure data storage and user authentication.
- 3) Victory Native and Skia libraries were integrated to design interactive data visualizations, offering non-technical users an intuitive platform to access predictive insights.

Phase 4: Testing and Performance Analysis

- 1) Stress tests were conducted on low-end devices to ensure system responsiveness, even in resource-constrained environments.
- 2) Predictive accuracy and AI-generated recommendations were validated using sample datasets, ensuring the reliability of system outputs under different environmental conditions.
- 3) Iterative feedback loops allowed the development team to refine performance, optimize user experience, and address emerging challenges during testing.

4.2 System Design

The EnviroSense system integrates real-time environmental monitoring, predictive analytics, and actionable insights to address challenges in modern agriculture. The system's architecture is designed around edge computing, enabling local data processing and rapid response to changing field conditions. This approach minimizes reliance on cloud services, reducing operational costs and improving reliability in low-connectivity regions.

4.3 Physical Architecture

The system consists of three main components: sensors, edge computing devices, and mobile applications.

The system utilizes a range of sensors to collect real-time data on critical environmental parameters, including temperature, humidity, soil pH, air quality, rainfall, light intensity, and nutrient levels (NPK). These sensors provide the foundation for monitoring crop health and environmental conditions, offering accurate and timely data for decision-making.

Arduino microcontrollers serve as edge devices for localized data processing. These devices are chosen for their cost-effectiveness, modularity, and compatibility with multiple sensors, making them particularly suitable for small-scale agricultural applications.

The system is developed using the React Native mobile application framework, known for its cross-platform compatibility and ability to create robust, responsive interfaces tailored to diverse user needs [17]. A mobile application serves as the primary interface for farmers and system-related settings. The app displays sensor data, predictions, and actionable recommendations in an intuitive format. Offline functionality and role-based access ensure usability across diverse user groups, even in areas with limited connectivity. All the necessary computations are done at the edge after the data is transferred to the local mobile device.

4.4 Data Flow

The data flow within the system follows these steps:

- 1) Sensors collect environmental data.
- 2) Data is processed locally by Arduino microcontrollers.
- 3) Processed data is transmitted to the online cloud database.
- 4) Machine learning models analyze the data to generate predictions and recommendations.
- 5) Insights are displayed in the mobile app, with email alerts triggered for critical conditions.

This streamlined data flow ensures efficient monitoring and actionable insights in real-time. The multimodal system proposed by [18] integrates IoT and machine learning to monitor environmental conditions and predict crop health. This methodology mirrors EnviroSense's approach by combining IoT sensors with machine learning algorithms, enabling real-time data processing and actionable insights for farmers. Incorporating a multimodal approach enhances system adaptability, improving scalability and overall performance.

5. AI Integration

The EnviroSense system incorporates advanced machine learning models to address key agricultural tasks, including field condition classification, yield prediction, and identifying ideal planting times. In [19] Deep learning models such as Long short-term memory (LSTM) and convolutional neural network (CNN) enhance climate prediction by analyzing environmental data patterns, allowing for real-time adjustments to greenhouse conditions. Each model is carefully selected for its ability to handle agricultural data and deliver actionable insights, empowering farmers with data-driven decisions to enhance productivity and reduce risks. The integration of machine learning algorithms in IoT-equipped smart agriculture systems enables real-time data analysis and decision-making, improving crop management and productivity[20].

5.1 Model Selection

For this reason, Random Forest was chosen as the main model for classifying field conditions and predicting the yield as it can work with non-linear relationships, assess feature importance, and handle large data sets without overfitting. This makes it easier to generalize across different conditions and makes it more effective for handling complex agricultural data sets as well.

Logistic Regression and Linear Regression were used as control variables. The logistic regression model was employed as a control variable for classification problems because of its efficiency, simplicity, and ease of interpretation. Although it can be used as a reliable base, due to its linearity it is unable to capture the more complex relationships existing in the data. Linear Regression was used for yield prediction as a basic model to understand the relationship between environmental factors and crop yield. However, it fails to capture nonlinear relationships between variables which hinders its effectiveness, especially in situations where there are complex agricultural conditions.

The choice of Random Forest over these alternatives was because it performed well with multicollinearity, non-linearity, and missing data which are issues often encountered in agricultural data.

5.2 Preprocessing and Feature Engineering

To enhance model accuracy and ensure reliable performance, preprocessing, and feature engineering techniques were applied to address key challenges in agricultural datasets, including noisy sensor data, class imbalances, and temporal patterns.

Rolling averages were applied to sensor data, smoothing temperature and humidity readings over a 7-day period. This reduces noise from daily fluctuations and allows the model to focus on long-term trends, leading to more stable and consistent predictions. By minimizing the impact of short-term outliers, the model can better track environmental patterns that influence crop health and yield.

Trend analysis was incorporated by calculating the differences between consecutive sensor readings, enabling the model to detect abrupt changes in environmental conditions. This improves the system's ability to respond to rapid shifts, helping identify potential threats to crops early and allowing for preventive measures.

To address class imbalances in the dataset, the Synthetic Minority Oversampling Technique (SMOTE) was used. In many cases, rare but critical conditions such as drought or pest infestations were underrepresented. SMOTE generates synthetic samples for these minority classes, ensuring the model can accurately classify rare scenarios and improve overall prediction performance.

Categorical features, such as "ideal planting time," were encoded using Label Encoding to convert them into numerical values suitable for machine learning algorithms. This transformation ensures that non-numeric factors critical to crop health are incorporated into the model, enabling the system to make more comprehensive and informed predictions.

By integrating these techniques, the model effectively processes real-time sensor data, detects environmental anomalies, handles imbalanced datasets, and leverages categorical features to deliver accurate and actionable insights for farmers.

5.3 Implementation Details

The machine learning pipeline is developed using Python and seamlessly integrated with the database. Sensor data is preprocessed and stored in the database, which ensures real-time updates and smooth synchronization with the mobile application. Models were trained on synthetic and historical datasets, with hyperparameter tuning conducted for Random Forest to optimize the number of estimators, maximum depth,

and feature splits. Once trained, the models were deployed on a cloud database for real-time prediction tasks, enabling farmers to access insights through an intuitive mobile interface.

5.4 Challenges and Improvements

- 1) **Class Imbalance:** Addressing class imbalance with SMOTE significantly improved the model's classification performance, particularly for underrepresented categories.
- 2) **Computational Efficiency:** Lightweight preprocessing methods and efficient algorithms minimized latency in predictions, ensuring the mobile application remained responsive, even on low-end devices.

By integrating robust machine learning models and advanced preprocessing techniques, the AI component of EnviroSense provides farmers with precise and timely insights, enabling optimized agricultural practices and improved resource management. Despite significant advancements, AI-driven systems encounter challenges related to sensor accuracy, data security, and regulatory compliance, which limit adoption among small-scale farmers [19].

6. Model Evaluation and Performance Analysis

This section evaluates machine learning models used in the EnviroSense system, including field condition classification, yield prediction, and feature importance analysis. Comparative results between models are highlighted to showcase the strengths and limitations of each approach.

6.1 Field Condition Classification

The classification of field conditions is a critical task in identifying healthy and underperforming fields. To evaluate the effectiveness of the models used, standard classification metrics—accuracy, precision, recall, and F1-Score—were applied to Random Forest and Logistic Regression models. Accuracy, which measures the overall correctness of predictions, showed that the Random Forest model performed exceptionally well with an accuracy of 98%, meaning it correctly classified 98% of all fields. In comparison, Logistic Regression achieved an accuracy of 85%, reflecting its limitations in handling the complexities of agricultural datasets (see Fig. 1).

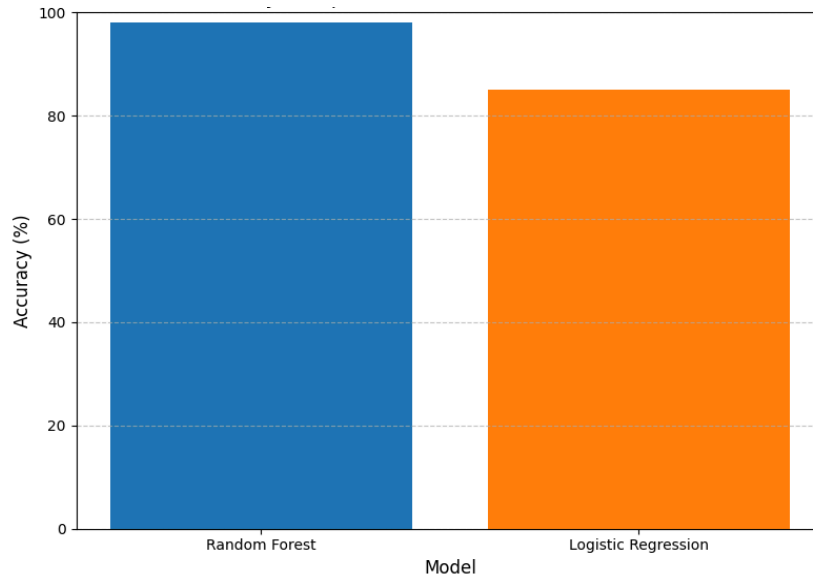


Fig.1. shows a bar chart comparing model accuracy.

Precision evaluates how often the model's predictions for a particular class were correct. For Random Forest, the precision for the "Healthy" class was 0.96, meaning 96% of the fields predicted as "Healthy" were truly healthy. For the "Needs Improvement" class, the precision was 0.98, demonstrating its ability to reliably identify problematic fields. In contrast, Logistic Regression struggled with the "Healthy" class, achieving a precision of only 0.44, which indicates a high rate of false positives. However, it performed better for the "Needs Improvement" class, with a precision of 0.98.

Recall, which measures the proportion of actual positives correctly identified, further highlights the strengths of the Random Forest model. For the "Healthy" class, Random Forest achieved a recall of 0.87, meaning it identified 87% of the truly "Healthy" fields, and for the "Needs Improvement" class, it achieved a perfect recall of 1.00, ensuring no problematic fields were missing. Logistic Regression matched Random Forest in recall for the "Healthy" class with a score of 0.87 but had a lower recall of 0.84 for the "Needs Improvement" class, indicating some underperforming fields were overlooked.

The F1-Score, which combines precision and recall into a single metric, provides a balanced view of the model's performance. For the Random Forest model, the F1-Score for the "Healthy" class was 0.91, reflecting a strong balance between precision and recall, while the "Needs Improvement" class achieved an F1-Score of 0.99, indicating near-perfect performance. Logistic Regression, however, had an F1-Score of only 0.58 for the "Healthy" class, reflecting its struggles with misclassification, and a stronger F1-Score of 0.90 for the "Needs Improvement" class, although still lower than Random Forest.

These results demonstrate the superiority of the Random Forest model in managing nonlinear relationships and imbalanced datasets, making it a more robust and reliable tool for field condition classification. While Logistic Regression is computationally efficient and provides acceptable results for the "Needs Improvement" class, its poor performance for the "Healthy" class highlights its inability to handle the complexities of agricultural data. The metrics underscore Random Forest's ability to provide actionable, accurate, and balanced insights for farmers, making it the preferred model for the EnviroSense system.

6.2 Yield Prediction

Accurate crop yield prediction is critical for effective planning and resource allocation in agriculture. To evaluate the performance of the models used in the EnviroSense system, the Random Forest and Linear Regression models were compared using Mean Squared Error (MSE), a metric that measures the average squared difference between actual and predicted values. Lower MSE values indicate more accurate predictions.

The Random Forest model achieved an MSE of 0.48, reflecting its ability to produce highly accurate predictions with minimal error. This low value indicates that the model effectively captured the complex, nonlinear relationships present in agricultural data, enabling reliable yield forecasting. In contrast, Linear Regression, serving as a baseline model, had a significantly higher MSE of 4.85, highlighting its limitations in handling complex data patterns. The higher MSE suggests that Linear Regression struggled to account for the nonlinear dependencies between environmental factors and crop yields, resulting in less accurate predictions (see Fig. 2).

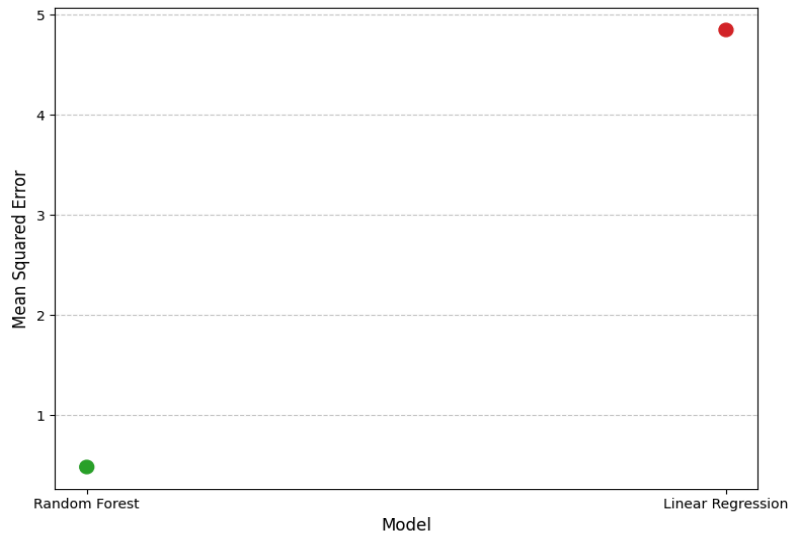


Fig.2. presents a scatter plot comparing the Mean Squared Error of the Random Forest model.

These results demonstrate the superiority of the Random Forest model for yield prediction tasks, particularly in diverse agricultural scenarios where data complexity is high. Its ability to minimize prediction errors and adapt to varying conditions underscores its suitability for resource allocation and planning, making it a valuable tool for enhancing productivity and sustainability in farming.

6.3 Feature Importance Analysis

To assess the key factors influencing the Random Forest model's predictions, feature importance analysis was conducted using explainable AI techniques, such as SHAP (SHapley Additive exPlanations). SHAP values help quantify the contribution of each feature to the model's output, providing insight into how different environmental variables affect predictions.

The analysis identified temperature, humidity, and soil pH as the most critical factors influencing crop health and yield predictions. These results align with established agricultural research, reinforcing the model's ability to capture essential environmental conditions that drive productivity. Secondary factors, including nutrient levels (NPK) and light intensity, were also shown to influence predictions, further refining the model's accuracy. (see Fig. 3)

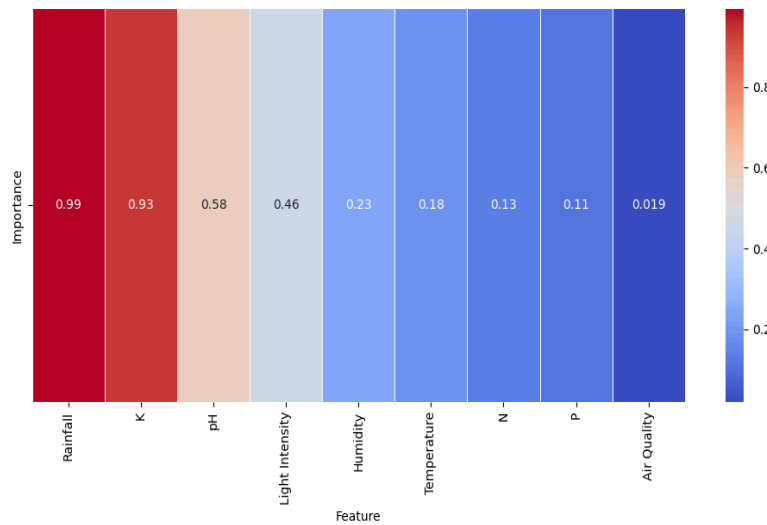


Fig.3. presents a heatmap illustrating the contribution of key environmental factors identified by the Random Forest model.

By using SHAP, the model's decisions can be explained in a simpler manner and therefore it becomes easier for the farmers to know the effect of each variable on the crop yield. For instance, change in humidity or soil pH has been seen to have a negative effect on the predictions which leads to measures like adjusting the irrigation or using soil treatments.

This approach not only proves the efficiency of the Random Forest model but also increases the confidence and the usefulness of the application by providing an insight

into how the predictions are reached. used The to application determine the best SHAP placement for feature resources importance and analysis to enhance yields the valuable well-being information of the crops. can be

Table 1. Summary of Model Performance Metrics

Task	Metric	Random Forest	Logistic Regression	Linear Regression
Field Condition Classification	Precision	Healthy: 0.96	Healthy: 0.44	N/A
		Needs Improvement: 0.98	Needs Improvement: 0.98	
	Recall	Healthy: 0.87	Healthy: 0.87	N/A
		Needs Improvement: 1.00	Needs Improvement: 0.84	
	F1-Score	Healthy: 0.91	Healthy: 0.58	N/A
		Needs Improvement: 0.99	Needs Improvement: 0.90	
	Overall Accuracy	0.98	0.85	
	Macro Avg F1-Score	0.95	0.74	
	Weighted Avg F1-Score	0.98	0.85	
	Mean Squared Error (MSE)	0.48	N/A	4.85
Ideal Planting Time Prediction	Precision	High across classes (e.g., "Early June": 0.96)	Lower for all categories	N/A
	Recall	High (e.g., "Early June": 0.95)	Lower for all categories	N/A
	F1-Score	High (e.g., "Early June": 0.95)	Lower for all categories	N/A
	Overall Accuracy	~96%	~84%	
Next Season Crop Prediction	Precision	High across classes (e.g., "Soybean": 0.97)	Lower across classes	N/A
	Recall	High (e.g., "Soybean": 0.96)	Lower across classes	N/A
	F1-Score	High (e.g., "Soybean": 0.96)	Lower across classes	N/A

	Overall Accuracy	~95%	~83%	
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The results highlight the superior accuracy of the Random Forest model compared to Logistic and Linear Regression models, particularly in complex classification tasks.

6.4 Discussion of Results

In this case, Random Forest gave the best results for the precision and recall which made it to possible determine the areas that require improvement accurately. The model had an accuracy of 98% and the precision and recall values for categories such as 'Healthy' and 'Needs Improvement' were greater than 0.90. These metrics provided a sound way of tracking the conditions of the fields and the appropriate recommendations. The second model, Logistic Regression served as a computationally efficient baseline and was unable to handle the complex data features and had an accuracy of 85% and low precision for the Healthy class.

Due to its capability of identifying non-linear relationships in the data set, Random Forest gave better results as compared to the Linear Regression model with an MSE of 0.48 and 4.85 respectively. This large difference in the prediction error shows that Random Forest is a better model for predicting yield in various agricultural conditions.

The study also established that temperature, humidity, and soil pH are the most crucial factors that affect crop health and yield as shown by the Random Forest model. These were identified as the most important factors and others include Nutrient levels (NPK) and light intensity in making the predictions. The importance of these features is in conformity with their effects on crop health and yield as evidenced by the model's predictions.

The experimental evidence confirms the effectiveness of the machine learning models incorporated in the EnviroSense system. Random Forest was determined to be the most stable model across all the tasks as it provided high levels of accuracy and reliable predictions. The information provided by these results helps to identify how to enhance the system's predictive and performance measures in the future.

7. Conclusion

The EnviroSense system integrates Artificial Intelligence (AI), the Internet of Things (IoT), and mobile edge computing into a unified platform that addresses critical challenges in modern agriculture. By providing farmers with actionable insights through real-time monitoring and predictive analytics, the system optimizes crop health, enhances yields, and reduces resource inefficiencies. AI-powered monitoring systems play a critical role in identifying crop diseases early and optimizing resource allocation, leading to improved productivity[21]. This solution bridges the gap between advanced agricultural technologies and small-scale farming, making precision agriculture more accessible and scalable.

Key outcomes of this study include the successful application of machine learning models, particularly Random Forest, which achieved an accuracy of 98% for field condition classification and outperformed baseline models in yield prediction tasks. Additionally, the development of a user-friendly mobile application ensures seamless

interaction with the system, enabling farmers to make data-driven decisions effectively.

The proposed approach demonstrates scalability and sustainability in farming practices through cutting-edge technologies. By aligning with global sustainability goals, such as the Sustainable Development Goals [SDGs], EnviroSense demonstrates a broader impact on enhancing food security and environmental resilience.

8. Future Work

Building upon the current implementation, several enhancements are planned to extend the system's capabilities and applicability:

- 1) Real-Time Sensor Data Integration:
 - a) The integration of MQTT protocols will enable live monitoring of environmental parameters, ensuring real-world validation of predictions and faster response times.
- 2) Deployment of Advanced Sensors:
 - a) Incorporating Arduino-based sensors will improve data collection in field conditions, capturing critical factors such as nutrient levels and rainfall with greater precision.
- 3) Seasonal Adaptation Algorithms:
 - a) Enhancing AI models to account for seasonal variations will improve the accuracy of predictions and recommendations, ensuring relevance across diverse climates.
- 4) Expansion to Additional Crops:
 - a) Adapting the system for a wider range of crops, such as barley and millet, will support broader agricultural applications and scalability.
- 5) AI Explainability:
 - a) Developing user-friendly explanations for AI-generated insights will build trust and usability for farmers, fostering greater adoption of the technology.

These future directions aim to refine EnviroSense's functionality and scalability, ensuring its adaptability to diverse agricultural needs and strengthening its impact on the farming ecosystem. The convergence of AI and IoT, known as AIoT, presents new opportunities for precision agriculture, enabling intelligent decision support systems and autonomous farming operations[22]. Future research will focus on scaling AI-driven climate control systems to larger farming operations to improve energy efficiency and reduce environmental impact [19].

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